

DAT 490 – Capstone Project (600 points)

Teams: You will need to complete the introduction in the Canvas discussion post and the Voicethread submission. You will be working in a team of 3-4 students. Once you have identified your team, please complete the Google spreadsheet in Canvas which will indicate that this has been completed.

Data: You will need to find a large dataset (>100K observations) that is of interest to you for this project. Here are some suggested websites that you can use:

<https://datahub.io/search>

<https://www.kaggle.com/>

<https://www.columnfivemedia.com/100-best-free-data-sources-infographic>

<https://rs.io/100-interesting-data-sets-for-statistics/>

<https://medium.com/analytics-vidhya/top-100-open-source-datasets-for-data-science-cd5a8d67cc3d>

Research Question(s): Develop 2-3 research questions that you can solve using an analytical approach to the dataset that you have selected for the project.

These components will be included in the Draft/Final report, but completed and submitted as separate assignments:

Project Plan (50 points) – Overview of your Capstone project which should include a description of the data, proposed research question(s) and proposed methodology.

Literature Review (50 points) – Internet search of research that has been completed related to your data.

Research Question(s) (25 points) – Identifies the research question(s) that you analyzed including reasons that you posed the question.

Exploratory Data Analysis (50 points) – Analysis of all variables that are used at the beginning of the analysis. Each variable should be graphed using histograms, scatterplots, boxplots with descriptive statistics.

Methodology (50 points)– What statistical procedures did you use to analyze the data.

Data Visualizations (50 points) – Graphs that were created to support the objectives of the analysis and/or helped to develop conclusions about the data.

Ethical Recommendations and Implications (50 points) – Explanation of the implications of your research which will include recommendations that are based on an ethical approach.

Grading note: If you do not receive full credit for any component, you can recover 50% of the deducted points once you make the necessary changes and resubmit the component within three days of when you receive the comments.

The Rough Draft will also include the following additional components:

Abstract/Executive Summary (25 points) – High level summary of the purpose, data, and summary of your findings. This should include some information about the company that you have chosen as your dependent variable.

Analysis (100 points) – This should include: analysis and results of each step, what decisions you made to proceed to the next step and why. All graphs/tables should be included and referenced, preferably in an Appendix. All source code should also be included in the Appendix, not in the body of the report.

Challenges (25 points) – Describing any challenges that you faced when collecting the data or performing the statistical analysis. How might they be mitigated in a future analysis?

Recommendations and Next Steps (25 points) – What were your overall conclusions from this analysis? Were there different steps that you would have taken or different data that you would have used if you were to complete this analysis again?

References (25 points) – Listing of all references using APA format. Any graph that was included in the report that was not developed by the student should be referenced as well.

The Final Report must include all changes/modifications to the Rough Draft (75 points).

Submission Guidelines – The Rough Draft and Final Report will be submitted including all source code and graphs in a PDF to be uploaded to Canvas.

Rough Draft Deadline: **Sunday, June 21 at 11:59 PM**

Final Report Deadline: **Friday, June 26 at 11:59 PM**